MIL-D-48120A(AR)
6 March 1978
SUPERSEDING
MIL-D-48120(PA)
4 March 1974

MILITARY SPECIFICATION

DETONATOR, ELECTRIC, M100

PARTS FOR
AND
LOADING, ASSEMBLING AND PACKING

*This specification is approved for use by the U.S. Army Armament Research and Development Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 This specification covers requirements, quality assurance provisions, packaging, fabrication, manufacture and loading of one type of electric detonator designated as Detonator, Electric, M100.

2. APPLICABLE DOCUMENTS

* 2.1 Issues of documents. - The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

TT-I-1795 - Ink, marking Stencil, Opaque for Porous Surfaces

MILITARY

MIL-P-116 - Preservation, Packaging, Methods of
MIL-B-117 - Bag, Sleeve and Tubing, Interior Packing
MIL-G-45204 - Gold Plating, Electrodeposited
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification For

FSC: 1375

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts
MIL-STD-331 - Fuze and Fuze Components, Environmental and performance tests for
MIL-STD-650 - Explosive Sampling, Inspection and Testing

DRAWINGS (see 6.6)

U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

8796522 - Marking Diagram and Sealing for Wood packing Boxes
9245691 - Detonator, Electric, M100
9298464 - S & A Assembly
9298456 - Lead Cup Assembly
9326705 - Microdetonator Clip Assembly

* 2.2 Other publications. - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transportation, Parts 100-199

(The Code of Federal Regulations is available form the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders should specify, "49 CFR 100-199 (latest revision)").

## 3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with the applicable drawings and specifications.

3.2 Detonators. - The detonators shall comply with all requirements specified on Drawing (Dwg) 9245691, all associated drawings, and with all requirements specified in applicable specifications.



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3.3 Moisture content

3.3.1 Lead azide. - The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.30 percent.

3.3.2 HMX. - The moisture content of the HMX at the loading station at the time of loading the detonators shall not exceed 0.30 percent.

* 3.4 Functioning output. - (The specific Functioning Output Requirements shall be determined by contract, see para. 6.2).

* 3.4.1 Lead cup functioning. - The detonator shall not dud, shall fire on the first attempt and produce a high order output of lead cup assembly 9298456 as defined by enlargement of the lead cup cavity in the S & A such that a 0.144 inch diameter gage plug can be freely inserted when tested as specified in para. 4.5.2.1.

3.4.2 Dent test functioning. - The detonator shall not dud, shall fire on the first attempt, and shall make a minimum depression of 0.005 inches in the steel witness plate when tested as specified in para. 4.5.2.2.

3.5 Waterproofness. - The detonators shall be waterproof as evidenced by normal functioning after being immersed in water maintained at 75 degrees Fahrenheit ($^{o}$F) plus or minus 5$^{o}$F, at a depth of 2 inches minimum, for not less than 48 hours.

3.6 Environmental

3.6.1 Resistance between electrode and cup, thermal shock and vibration. - The detonators shall comply with the resistance requirement on Dwg. 9245691 and with output requirements of para. 3.4.2 when exposed to thermal shock and vibration.

3.6.2 Cold temperature. - The detonator assembly shall comply with output requirements of para. 3.4.2 when conditioned at -65 $\pm$ 5$^{o}$F for two hours.

3.6.3 Hot temperature. - The detonator assembly shall comply with output requirements of para. 3.4.2 when conditioned at +165 $\pm$ 5$^{o}$F for two hours.

* 3.7 Shorting during shipping. - The detonator shall be shorted when assembled into the microdetonator clip assembly (Dwg 9326705) as evidenced by a resistance of 0.5 ohm maximum between the detonator electrode and the cup.



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3.8 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.9 Workmanship. - All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. Cleaning methods used shall not be injurious to any parts, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

## 4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

* 4.2 Classification of inspections. - The following types of inspection shall be conducted on this item:

a. First Article Inspection
b. Quality Conformance Inspection

* 4.3 First article inspection. -

4.3.1 Submission. - The contractor shall submit a first article as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.2. The first article sample shall consist of the following items in sample quantities as indicated.

| Part Description | Drawings | Quantity |
|---|---|---|
| Cup | 9245697 | 50 |
| Plug Assembly | 9245694 | 50 |
| Bridge Plug and Ignition Assembly | 9245692 | 50 |
| Charged Cup | 9257046 | 50 |
| Detonator Assembly | 9245691 | 255 or 355 See Note |

NOTE: The quantity of detonators required shall be in accordance with the output requirements specified in the contract. If the dent test output requirement is specified 355 samples shall be submitted. If the lead cup output requirement is specified 255 samples shall be submitted.



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4.3.2 Inspections to be performed. - See MIL-A-48078 and Table I herein.

4.3.3 Rejection. - See MIL-A-48078, and Table I herein.



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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|---|---|
| | Detonator Assembly and Components | | | | See below |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | - |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| | Cup (Dwg. 9245697) | | | | |
| | Examination for defects | 50 | | 3.2 | 4.4.2.1 |
| | Plug Assembly (Dwg. 9245694) | | | | |
| | Examination for defects | 50 | | 3.2 | 4.4.2.3 |
| | Pull test of electrode | 50 | | 3.2 | 4.5.4 |
| | Insulation resistance between electrode and header | 50 | | 3.2 | 4.5.5 |
| | Bridge Plug and Ignition Assembly (Dwg. 9245692) | | | | |
| | Examination for defects | 50 | | 3.2 | 4.4.2.4 |
| | Detonator Assembly (Dwg. 9245691) | | | | |
| | Examination for defects | Note 1 | | 3.2 | 4.4.2.6 |
| | Waterproofness | 80 | 80-0-1 | 3.5 | 4.5.2.1 |
| | Functioning output (dent test) when applicable | 200 | 200-0-1 | 3.4.2 | |
| | Functioning ouput (lead cup) when applicable | 100 | 100-1-2 | 3.4.1 | |
| | Environmental tests | | | | |
| | (a) Resistance between electrode and cup, thermal shock and vibration | 25 | 25-0-1 | 3.6.1 | 4.5.3.1 |
| | (b) Cold temperature conditioning | 25 | 25-0-1 | 3.6.2 | 4.5.3.2 |
| | (c) Hot temperature conditioning | 25 | 25-0-1 | 3.6.3 | 4.5.3.3 |

NOTES: 1. The entire quantity of detonators required for the tests below shall be subjected to the examination for defects prior to testing.

4.4 Quality conformance inspection

* 4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of electric detonators shall contain:

a. Component parts from one lot interfix number from one manufacturer.

b. HMX from one lot number only

c. Lead azide of the same type from not more than two lots from the same manufacturer, in accordance with the same specification and specification revision and which have subsequently been subjected to uniformly controlled drying and processing conditions by the manufacturer. If more than one lead azide lot is used the lots shall not differ by more than three serial lot numbers.

* 4.4.2 Examination. - See MIL-A-48078.

a. Sampling plans. - Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.



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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.1 | Cup | | 9245697 |
| | | | NEXT HIGHER ASSEMBLY |
| | | | 9245691 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Thickness through bottom | | 0.40% | 3.2 | Gage |
| 102 | Outside diameter | | 0.40% | 3.2 | Gage |
| 103 | Concentricity of outside diameter with inside diameter | | 0.40% | 3.2 | Gage |
| 104 | Spherical radius | | 0.40% | 3.2 | Gage |
| 105 | Dented or distorted | | 0.40% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Inside diameter, minimum | | 0.65% | 3.2 | Gage |
| 202 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

NOTES:

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| *4.4.2.2 | Bridgewire | | | | 9326203 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9245693 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Moleculoy wire dia | | See Note | 3.2 | Gage |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

NOTES: Five (5) measurements shall be made in the first two (2) feet of wire on each spool, and at least once per shift thereafter.





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QUALITY CONFORMANCE INSPECTION

# CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | SHEET 1 OF 1 | DRAWING NUMBER |
|---|---|---|---|
| 4.4.2.3 | Plug Assembly | | 9245694 |
| | | | NEXT HIGHER ASSEMBLY |
| | | | 9245691 |

| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
|---|---|---|---|---|---|
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pull test of electrode | 50 | 50-0-1 | 3.2 | 4.5.4 |
| 102 | Insulation resistance test | | 100% | 3.2 | 4.5.5 |
| 103 | Outside diameter, max. | | 0.40% | 3.2 | Gage |
| 104 | Bead cracked | | 0.40% | 3.2 | Visual |
| 105 | Gold plating | Note 1 | Note 1 | 3.2 | 4.5.8 |
| Minor | | | | | |
| 201 | Length of electrode below surface of header | | 0.65% | 3.2 | Gage |
| 202 | Thickness of header | | 0.65% | 3.2 | Gage |
| 203 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

NOTES:
1. Sampling and rejection shall be in accordance with MIL-G-45204.



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## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.4 | Bridge Plug and Ignition Assembly | | | | 9245692 |
| | | | | | NEXT HIGHER ASSEMBLY 9245691 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Spot charge missing | | 0.40% | 3.2 | Visual |
| 102 | Cracks or voids in spot charge | | 0.40% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.5 | Charged Cup | | | | 9257046 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9245691 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Height of total charge | | 0.40% | 3.2 | Gage |
| 102 | Moisture Content of Lead Azide | | 4.4.3.1 | 3.3.1 | 4.5.1.1 |
| 103 | Moisture Content of HMX | | 4.4.3.1 | 3.3.2 | 4.5.1.2 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

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NOTES:



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QUALITY CONFORMANCE INSPECTION

## CLASSIFICATION OF DEFECTS & TESTS

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.6 | Detonator Assembly | | | | 9245691 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | - |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | | | | | |
| 1 | Cup cracked or split | | 100% | 3.2 | Visual |
| Major | | | | | |
| 101 | Length of electrode | | 0.40% | 3.2 | Gage |
| 102 | Length, crimped end to base | | 0.40% | 3.2 | Gage |
| 103 | Depth to spherical radius | | 0.40% | 3.2 | Gage |
| 104 | Large diameter | | 0.40% | 3.2 | Gage |
| 105 | Spherical radius missing or incorrect | | 0.40% | 3.2 | Gage |
| 106 | Spherical radius scratched, damaged or dented | | 0.40% | 3.2 | Visual |
| 107 | Sealant not present 360° | | 0.40% | 3.2 | Visual |
| 108 | Presence of sealant on electrode beyond sealed area | | 0.40% | 3.2 | Visual |
| 109 | Waterproofness test | 50 | 0-1Note1 | 3.5 | 4.5.2.2 |
| 110 | Functioning output (dent test) | 200 | 1-2Note1 | 3.4.2 | |
| 111 | Functioning output (lead cup test) | 80 | 1-2Note1 | 3.4.1 | |
| 112 | Environmental tests | | | | |
| | (a) Resistance between electrode and cup | | 100% | 3.2 | 4.5.3.1 |
| | (b) Cold temperature conditioning | 25 | 0-1 | 3.2 | 4.5.3.2 |
| | (c) Hot temperature conditioning | 25 | 0-1 | 3.2 | 4.5.3.3 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

NOTES: 1. Any detonator that fails to function on the first attempt or fails to produce the required output shall be classed as a defective.

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QUALITY CONFORMANCE INSPECTION

**CLASSIFICATION OF DEFECTS & TESTS**

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| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.7 | Microdetonator cup assemlby, (after assembly of detonator) | | | | 9326705 |
| | | | | | NEXT HIGHER ASSEMBLY |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | | | | | |
| 1 | Detonator not shorted | | 100% | 3.7 | Current limited ohmmeter (1 milliamp, max.) |
| Major | | | | | |
| 101 | Detonator electrode protruding above flush of clip assembly | | 100% | 3.7 | Visual |
| 102 | Detonator cup protruding above flush of clip assembly | | 100% | 3.7 | Visual |
| 103 | Detonator inverted in clip | | 100% | 3.7 | Visual |
| 104 | Spherical radius scratched, damaged or dented | | 100% | 3.2 | Visual |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

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## CLASSIFICATION OF DEFECTS & TESTS



| PARAGRAPH | TITLE | | SHEET 1 OF 1 | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.8 | Sealed Bag | | | | 8796522 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | - |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Heat seal test | Note 1 | Note 1 | 3.2 | 4.5.9 |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | 1.0% | 3.9 | Visual |

NOTES: 1. Sampling and rejection shall be in accordance with MIL-P-116.





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* 4.4.3 Testing

4.4.3.1 Moisture content. -

TABLE I

| Material | Requirement | Test Paragraph |
|---|---|---|
| Lead azide | 3.3.1 | 4.5.1.1 |
| HMX | 3.3.2 | 4.5.1.2 |

The contractor shall provide adequate controls to insure compliance with the requirements and shall test for verification one (1) sample of each material from each eight (8) hours of detonator production. Composite samples shall not be used. If the moisture content of a sample exceeds the requirements and loading has not begun, that quantity of material represented by the sample shall be rejected. If detonators have been loaded with material containing excessive moisture, the remaining material that is represented by the sample, together with all detonators loaded with the non-conforming material, shall be rejected.

* 4.4.4 Inspection equipment. - The inspection equipment required to perform the examinations and tests prescribed herein is described in the Paragraph Reference/Inspection Method column in the tables starting with Paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.3 herein.

* 4.5 Test methods and procedures. -

4.5.1 Moisture content. -

4.5.1.1 Lead azide. - The moisture content of the lead azide shall be determined in accordance with Method 101.5 of MIL-STD-650 except that a two (2) gram sample shall be tested and the dish shall be covered with a sheet of polytetrafluoroethylene.

4.5.1.2 HMX. - The moisture content of HMX shall be determined as specified in 4.5.1.1.

4.5.2 Functioning output tests

* 4.5.2.1 Functioning output (lead cup). - The detonator and lead cup shall be assembled into an armed S & A (Dwg. 9298464) and shall be temperature conditioned at 70 $\pm$ 10°



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degrees Fahrenheit ($^{o}$F) for two (2) hours minimum prior to performing the functioning tests. The detonator shall be fired by applying the energy from a 100 microfarad capacitor charged to 1.6 volts maximum. Observation shall be made for any failure to comply with the requirements specified in 3.4.1.

4.5.2.2 Functioning output (steel plate). - The detonator assemblies shall be temperature conditioned at 70 ± 10 degrees Fahrenheit ($^{o}$F) for two (2) hours minimum prior to performing the functioning tests. The tests shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate be 1/8 inch minimum and optionally round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and the other to the electrode. The detonator shall be fired by applying the energy from a 100 maximum microfarad capacitor charged to 1.6 volts maximum. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of para. 3.4.2.

4.5.2.3 Waterproofness output. - The detonator assemblies shall be immersed in water at a two (2) to three (3) inch depth for 48 hours minimum. The water temperature shall be maintained at 70 ± 10$^{o}$F for the immersion period. At the end of the immersion time, the detonator assemblies shall be removed from the water, assembled into the test fixture and fired by applying the energy from a 100 maximum microfarad capacitor charged to 1.6 volts maximum. The depression in the witness block shall be measured using a dial indicator equipped with a probe having a 0.003 ± 0.001 inch spherical radius and in accordance with the requirements of para. 3.4.2.

The test shall be performed as specified in MIL-STD-331, Test 301.1 except that the witness plate be 1/8 inch minimum and optionally round, 1.22 - .01 dia. Witness plate hardness shall be Rockwell B70 to B95. The base of the detonator shall be placed directly against the center of the witness plate and electrical contact made through a clip which will provide a minimum confinement of the detonator. One lead shall be affixed to the detonator cup and other to the electrode.

4.5.3 Environmental tests

4.5.3.1 Resistance between electrode and cup, thermal shock, and vibration. - The detonator assemblies shall be tested in the following sequence.



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a. Measure resistance between the electrode and cup in accordance with 4.5.7.

b. Condition for two (2) hours minimum as per para 3.6.2.

c. Remove from cold chamber and immediately transfer to the hot chamber.

d. Condition for two (2) hours minimum as per para 3.6.3.

e. Remove from hot chamber and immediately transfer to the vibrator.

f. Vibrate for 15 minutes minimum in a sweep covering frequency range of 25 - 500 - 25 cycles per second parallel to the transverse orthogonal axis with base charge down and a 2 ± 0.2 g's peak.

g. Measure resistance between electrode and cup in accordance with 4.5.7.

h. The detonators shall be functioned as specified in 4.5.2.1 or 4.5.2.2.

4.5.3.2 Cold temperature. - The detonator assemblies shall be conditioned to -65 ± 5°F for two (2) hours minimum. The detonator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.2.1 or 4.5.2.2.

4.5.3.3 Hot temperature. - The detonator assemblies shall be conditioned to +165 ± 5°F for two (2) hours minimum. The detonator assemblies, maintained at the conditioning temperature, shall be functioned as specified in 4.5.2.1 or 4.5.2.2.

4.5.4 Pull test of electrode. - The plug assembly shall be assembled to the equipment and tested in accordance with Method 211 of MIL-STD-202. The electrode and header shall be observed for compliance with the requirement. This is a destructive test. Test samples shall not be returned to the lot.

4.5.5 Insulation resistance between the electrode and header. - The plug assembly shall be connected to the equipment and the required voltage applied for not less than one (1) nor more than two (2) seconds. The equipment shall be observed to determine compliance with the requirements. Any plug assembly that fails to comply with the requirements shall be classed defective and removed from the lot.



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4.5.6 Resistance between electrode and header. - The bridge assembly shall be connected to the equipment and the bridge resistance shall be measured when a current limited to one (1) milliampere is appled. Any bridge assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.7 Resistance between electrode and cup. - The detonator assembly shall be connected to the equipment and the resistance shall be measured between the electrode and the cup when a current limited to one (1) milliampere is applied. Any detonator assembly that fails to comply with the requirements shall be classed defective and removed from the lot.

4.5.8 Gold plating of pin in plug assembly (see Dwg 9245694 covering a detail of Dwg 9245691). - Sampling, acceptance criteria and test procedures for this test shall be as specified in Specification MIL-G-45204.

4.5.9 Heat seal test of sealed bag (see Dwg 8796522). - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

## 5. PREPARATION FOR DELIVERY

* 5.1 Preservation and packaging. - The electric detonators shall be packaged in accordance with Code of Federal Regulations Title 49 Parts 100-199. In addition, each detonator shall be electrically shorted during shipment and storage by means of a metallic connection from the detonator pin to the cup by assembly into the microdetonator clip assembly, Dwg. No. 9326705.

Each intermediate carton, set-up box or comparable container, after blunting corners, shall be placed in a water and vapor proof bag manufactured in accordance with Type I, Class E of Specification MIL-B-117. The bag shall be closed. All excess air shall be exhausted within the bag and the bag then sealed in accordance with the manufacturer's instructions. The seal shall be located at the edge of the bag to provide for a minimum of two (2) subsequent seals, if required. The seal shall comply with the heat seal requirements of Specification MIL-P-116.

* 5.2 Packing. - A suitable number of packaged detonators shall be packed in an outer box in accordance with Code of Federal Regulations Title 49, Parts 100-199.

5.3 Marking. - Each inner container and intermediate carton set-up box or comparable container shall be printed,



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stamped or labeled and each barrier bag printed or stamped in accordance with the following information and instructions:

a. (Number) Detonator, Electric, M100
Lot No. Date Packed (mo/Yr)

b. All letters shall be upper case. Size of letters and numerals shall be a minimum of 1/4 inch high applied with black waterproof ink. Stencil ink shall conform to Specification TT-I-1795.

c. Printing and labeling shall be commercial type. Labels shall be on white gummed paper.

d. Exterior boxes shall be marked in accordance with Dwg 8796522. The DOT designation shall be: Electric Blasting Caps - Handle Carefully (see Note).

e. Note: "The number of caps packed in one (1) outer pack shall be shown in the marking."

* 5.4 Shipping. - When detonator from more than one lot are shipped at one time, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

* 6.1 Intended use. - The detonator assembly covered by the specification is intended for use in munition assemblies.

* 6.2 Ordering data. - See Sections 6.1 and 6.2 of MIL-A-48078. In addition the contract shall specify the applicable output requirements of 3.4 based on the intended use of the electric detonator.

6.3 Changes from previous issue. - The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

6.4 The Government as referred to in this specification is the technical agency listed in 6.6.



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6.5 Submission of inspection equipment designs for approval. - (See MIL-A-48078). - Submit equipment designs as required to:

Commander
ARRADCOM
ATTN: DRDAR-QAA-I
Dover, New Jersey 07801

6.6 Distribution of ammunition data cards, inspection and test results, process controls. - Distribution of ammunition data cards, inspection and test results, process controls shall be sent to:

Commander
ARRADCOM
ATTN: DRDAR-QAA-Q
Dover, New Jersey 07801

* 6.7 Drawings. - Drawings listed in Section 2 of this specification under the heading U.S. Army Armament Research and Development Command (ARRADCOM) may also include drawings prepared by, and identified as, Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARRADCOM.

Custodian:
Army-AR

Preparing Activity:
Army-AR

Project Number: 1375-A-214



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MIL-D-48120A Detonator, Electric, M100 Parts for & LAP

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☐ VENDOR ☐ USER ☐ MANUFACTURER

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